

# REAGAN HU

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## EDUCATION

### University of Toronto - St. George Campus

Engineering Science - Bachelors of Applied Science and Engineering (Co-op) GPA: 3.8/4.0

University of Toronto Scholars Scholarship (\$10000)

## EXPERIENCE

### Simulation Team Engineer | aUToronto

August 2026 - Present

- Developed a **trajectory pipeline** to convert raw path data into timestamped motion references for control simulations.
- Built a local web GUI for simulation config selection to improve start-up efficiency while introducing pre-run validation.

### AI Robotics Engineer Intern | Acceleration Consortium

May 2026 – August 2026

- Engineered automated **Isaac Sim digital twins** allowing repeatable, high-volume data collection of **5000+ demonstrations** for **imitation learning** and **reinforcement learning** enabling scalable policy training and investor-facing demonstrations.
- Established the first company-wide **single-operator** physical data collection workflow by developing an internal tool for voice-assisted teleoperation (**faster-whisper**) and automated segmentation for data annotation (**CV2, SAM2, PyTorch**).
- Built an end-to-end inference pipeline (**WebSocket, PyZMQ, PyTorch**) enabling autonomous deployment of fine-tuned **VLA** policies supporting **DSRL** and **RTC** for multi-step workflows showcased to multi-billion-dollar venture capital firms.
- Increased precision autonomous manipulation accuracy by **68%** by developing an end-to-end, safety-gated visual-servo pipeline (**MolmoPoint, SciPy, ZED SDK**) for reliable closed-loop robot alignment.

### Autonomous Rover Embedded Engineer | University of Toronto Robotics Association

September 2025 - April 2026

- Improved rover fault monitor execution throughput by **7.7x** through refactoring embedded **C++** codebase refactor into modular **FreeRTOS** task-based architecture with deterministic scheduling and **mutex**-protected shared variables.
- Reduced rover operation time by **67%** through building a **PyQt5** control GUI on **ROS-2 Humble**, implementing publisher-subscriber pipelines for live node/topic monitoring, launch control, and emergency stop handling.
- Developed a **Python** GPS/IMU analysis pipeline (**NumPy, Pandas, Matplotlib, PyProj**) transforming LLA to ENU coordinates with **Jacobian-based** covariance propagation for performance evaluation.

## PROJECTS

### AroundU 🗣️

*Skills & Tools:* React, TypeScript, TailwindCSS, Vite, Motion, Node.js, Express, Supabase, OpenAI API, Hugging Face

- Built a full-stack campus social platform (**React, TypeScript, Vite, TailwindCSS**), implementing onboarding flows, real-time event discovery, messaging, and an interactive heatmap with **Google Maps API**.
- Developed reusable, animated interface components (**Motion, Lucide React**) supporting navigation and live visualization.
- Integrated a **Python** recommendation pipeline using **Hugging Face sentence-transformers** to embed and compare user profiles, with **OpenAI API** extracting and weighting data to generate compatibility scores.
- Contributed to backend development using **Node.js, Express, and Supabase PostgreSQL**, supporting user profiles, events, chats, and interest-based matching.

### Fast and Flurrious Autonomous Rover 🗣️

*Skills & Tools:* C++, Sensor Integration, PID control, PWM, Arduino

- Implemented a real-time line-following controller using calibrated **TCS3200 color-sensor** readings and tuned **PID control**, dynamically adjusting differential motor speeds through **PWM** for precise path tracking and line-loss recovery.
- Integrated an **ultrasonic distance sensor, L298N motor driver**, and servo gripper for autonomous obstacle avoidance and object pickup and placement execution.

## VOLUNTEERING

### National Director of Design | Hot Potato Initiative

September 2024 - June 2025

*Skills:* Marketing Design, Professional Collaboration, Project Management

- Coordinated a team of **10** in the development marketing materials for nationwide campaigns leading to **35%** outreach growth.
- Directed end-to-end creative workflows to develop national campaigns that balanced needs stakeholder needs with advocacy.

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, C++, C, Matlab

**Tools & Libraries:** Isaac Sim, CloudXR, SAM, PyTorch, TensorFlow, JAX, OpenCV, FreeRTOS, ROS-2, Pandas, Numpy

**Skills:** Forward/Inverse Kinematics, Digital Twins, VLA Fine-Tuning, Visual Servo, SSH, Git, Linux

**Coursework:** CAD (Onshape), Circuit Design, Linear Algebra, Technical Writing

**Languages:** English, French, Mandarin